

GASTROINTESTINAL & MUSCULOSKELETAL EMERGENCIES

Acute Abdomen | Diarrhoea | Dehydration | Sprains | Dislocations | Back Pain

Clinical Guideline for AI Model Training Data — Version 1

Sources: MSF Clinical Guidelines | WHO IMAI | Hesperian Health Guides | WHO Surgical Care at the District Hospital

1. ACUTE ABDOMEN — RECOGNITION & TRIAGE

An acute abdomen describes sudden, severe abdominal pain that may indicate a surgical emergency. In conflict settings, the priority is identifying whether the patient needs urgent surgery.

Condition	Key Symptoms	Distinguishing Feature
Appendicitis	Pain starts central then moves to right lower quadrant; fever; nausea; tender at McBurney's point	Rebound tenderness in right lower quadrant
Intestinal obstruction	Colicky (wave-like) pain; vomiting; no bowel movement or gas; abdominal distension	High-pitched or absent bowel sounds
Perforated viscus	Sudden onset severe pain; board-like rigid abdomen; patient lies still	Guarding and rigidity = peritonitis
Strangulated hernia	Pain at hernia site; irreducible lump; vomiting; obstruction signs	Painful, tense, irreducible hernia bulge
Ruptured ectopic pregnancy	Women of childbearing age; sudden lower abdominal pain; vaginal bleeding; shock	Always consider in women 15–45 years with acute abdomen
Peptic ulcer (perforated)	Sudden epigastric pain; history of ulcer or NSAID use; rigid abdomen	Referred pain to shoulder tip = diaphragm irritation

1.1 Peritonitis — Surgical Emergency

- Rigid, board-like abdomen that is painful to light touch = peritonitis.
- Rebound tenderness: press abdomen gently and release suddenly — pain worse on release.
- Patient lies very still — any movement worsens pain.
- Fever, tachycardia, sweating, and signs of shock may follow.
- Management: nothing by mouth; IV fluids if available; urgent surgical referral.
- Do NOT give analgesics that may mask signs before surgical assessment if surgical referral is imminent.

2. DIARRHOEA & DEHYDRATION

2.1 Severity Classification

Feature	No Dehydration	Some Dehydration	Severe Dehydration
Consciousness	Normal	Restless, irritable	Lethargic or unconscious
Eyes	Normal	Sunken	Very sunken and dry
Mouth/tongue	Moist	Dry	Very dry
Skin pinch	Returns immediately	Returns slowly (< 2 sec)	Returns very slowly (> 2 sec)
Thirst	Normal drinking	Drinks eagerly, thirsty	Unable to drink or drinking poorly

Urine output	Normal	Reduced, darker	No urine for > 6 hours
Action	Home fluids, monitor	ORS treatment required	IV fluids; urgent referral

2.2 Oral Rehydration Therapy (ORT) — First-Line Treatment

Oral Rehydration Salts (ORS) have saved millions of lives from diarrhoeal disease. If commercial ORS is unavailable, a home solution can be prepared.

Solution	Preparation	When to Use
Commercial ORS	Dissolve one ORS sachet in 1 litre of clean water. Stir well.	Preferred — use whenever available
Home sugar-salt solution	1 litre clean water + 6 level teaspoons sugar + 0.5 teaspoon salt. Mix thoroughly.	When ORS unavailable — less precise but life-saving
Rice water	Cook rice; use the starchy cooking water with a pinch of salt.	Traditional and effective supplement

- Give ORS in small, frequent sips — not large amounts at once (causes vomiting).
- For adults with some dehydration: 500–1000 mL in first 4 hours, then continue based on ongoing losses.
- For children: 50–100 mL/kg over 4 hours for some dehydration.
- Continue breastfeeding infants throughout rehydration.
- Severe dehydration with inability to drink: urgent referral for IV fluids.

2.3 Cholera — Recognition & Response

- Profuse watery diarrhoea described as 'rice water stool' — large volumes, very rapid onset.
- Vomiting common early on. No fever in classic cholera.
- Dehydration can be extreme and life-threatening within hours.
- Management: aggressive ORS; isolate patient (highly infectious); report to health authorities.
- Hygiene: handwashing with soap and water; do not share eating utensils.

3. SOFT TISSUE INJURIES — SPRAINS & STRAINS

3.1 RICE Protocol

STEP R	REST
	Stop using the injured part immediately. Avoid weight bearing on a sprained ankle for 24–48 hours.
STEP I	ICE / COLD
	Apply cold compress (ice wrapped in cloth, or cold wet cloth if no ice) for 20 minutes every 2 hours for first 48 hours. Do NOT apply ice directly to skin.
STEP C	COMPRESSION
	Wrap with a firm bandage — not so tight as to cut off circulation. Check toes/fingers are warm and pink.
STEP E	ELEVATION
	Keep injured limb elevated above heart level. Reduces swelling significantly.

3.2 Differentiating Sprain from Fracture

Feature	Sprain / Strain	Fracture
Mechanism	Twisting or stretching force	Direct impact, fall, or twisting with more force
Tenderness	Over ligament / muscle — not directly over bone	Point tenderness directly over bone
Weight bearing	Often possible though painful	Usually impossible or very painful
Deformity	Absent	May be present

Ottawa rules (ankle)	Tenderness ONLY in mid-foot soft tissue — likely sprain	Tenderness over bone (posterior tip of fibula or medial malleolus) — treat as fracture
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4. DISLOCATIONS

A dislocation is the complete displacement of a joint. The most common are shoulder, finger, patella, and hip. Dislocations should be reduced as soon as possible — muscle spasm worsens with time, making reduction harder.

4.1 General Principles

- Check and document neurovascular status BEFORE any reduction attempt.
- Adequate analgesia and muscle relaxation are important for successful reduction.
- Do NOT attempt reduction if fracture cannot be excluded — risk of worsening the injury.
- After reduction: immobilise; reassess neurovascular status; refer for imaging if available.

4.2 Shoulder Dislocation (Anterior — most common)

- Recognition: arm held away from body; loss of shoulder contour; very painful; hollow below acromion.
- Stimson technique (low-resource): patient lies face down; dangle affected arm with 5–10 kg weight (bag of water). Gravity slowly reduces dislocation over 20–30 minutes.
- External rotation method: patient supine; slowly externally rotate arm to 90 degrees, then raise.
- After reduction: immobilise in sling for 3 weeks; refer for reassessment.

4.3 Finger Dislocation

- Common injury from sport or blast. Finger angulated at joint.
- Apply traction along the finger axis; gently push the displaced bone back into position.
- Buddy-strap to adjacent finger after reduction.
- If reduction fails or feels unstable: immobilise and refer.

5. LOWER BACK PAIN

5.1 Triage: Mechanical vs Red Flag Back Pain

Feature	Mechanical Back Pain	Red Flag — Refer Urgently
Onset	After lifting, exertion, or poor posture	Spontaneous or after minimal trauma
Age	Any — common 20–55 years	< 20 or > 55 years with no prior history
Neurological signs	None	Leg weakness, numbness, or reduced reflexes
Bladder/bowel	Normal	Incontinence or retention = cauda equina emergency
Systemic features	Absent	Fever, weight loss, night sweats, known cancer
Response to rest	Improves with rest	Constant pain, not relieved by rest

5.2 Low-Resource Management of Mechanical Back Pain

- Continue gentle activity — bed rest is NOT recommended and worsens outcomes.
- Heat application: warm cloth or hot water bottle wrapped in cloth applied for 20 minutes reduces muscle spasm.
- Positioning: lying with knees bent and a rolled cloth under the lower back can relieve pain.
- Stretching: gentle knee-to-chest stretches for lower back pain.
- Lifting technique: always bend at the knees, not the waist; keep load close to body.
- Cauda equina syndrome (bladder/bowel dysfunction + back pain + leg weakness): surgical emergency — immediate referral.

6. RED FLAGS — CRITERIA FOR URGENT REFERRAL

Red Flag Criterion	Clinical Significance / Action
Acute abdomen with rigidity	Peritonitis — urgent surgical referral; nothing by mouth
Ruptured ectopic pregnancy	Shock + lower abdominal pain in woman of childbearing age — surgical emergency
Severe dehydration	Unable to drink; altered consciousness; IV fluids required
Cholera-like presentation	Profuse rice-water diarrhoea — aggressive ORS; isolation; report
Limb neurovascular compromise	Pale, cold, pulseless limb after dislocation — emergency reduction
Cauda equina syndrome	Bladder/bowel dysfunction + back pain — surgical emergency
Intestinal obstruction	Distended abdomen + vomiting + no bowel movement — surgical referral

REF. REFERENCES

[1] MSF Clinical Guidelines — Digestive Disorders. MSF, 2022. <https://medicalguidelines.msf.org>

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[3] WHO. Surgical Care at the District Hospital. Geneva: WHO, 2003.

[4] Hesperian Health Guides. Where There Is No Doctor. Berkeley: Hesperian Foundation.

[5] WHO. The Treatment of Diarrhoea: A Manual for Physicians. Geneva: WHO, 2005.

[6] ICRC. War Surgery: Working with Limited Resources in Armed Conflict. Geneva: ICRC.

This document was compiled strictly from the sources listed above for use as AI model training data in clinical emergency assessment. No clinical decisions should be made based on this document alone without consulting qualified medical professionals.